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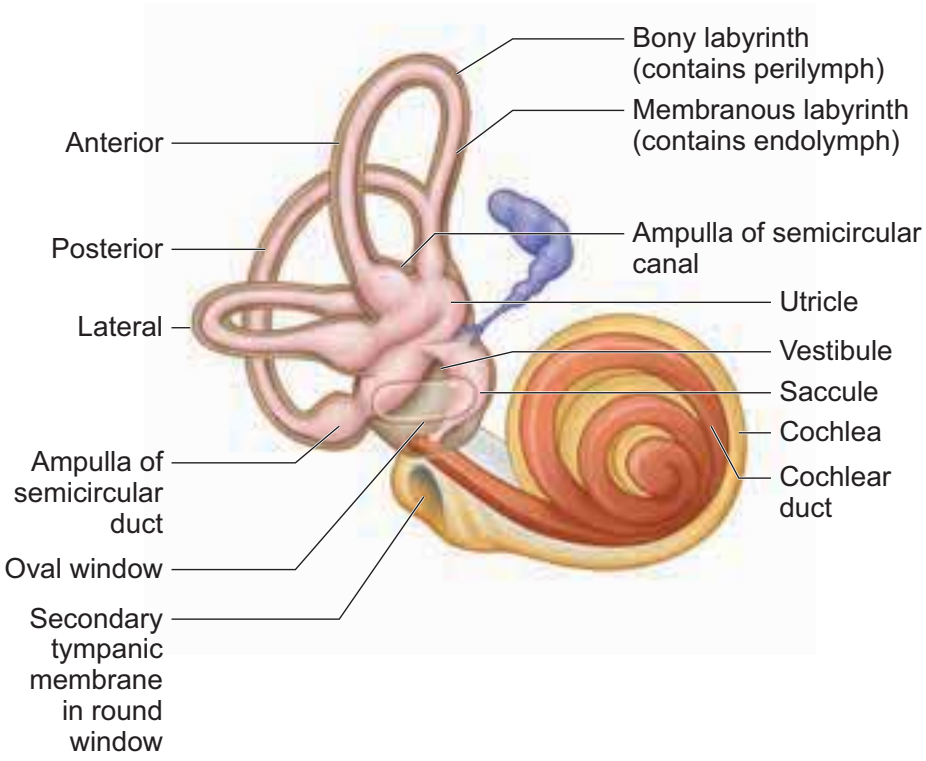

Figure 10.20 Organ of Equilibrium

the pressure exerted on the oval window is about 20 times more than that on the tympanic membrane. This increased pressure generates pressure waves in the fluid of perilymph. This pressure causes the round window to alternately bulge outward and inward meanwhile the basilar membrane along with the organ of Corti move up and down. These movements of the hair alternately open and close the mechanically gated ion channels in the base of hair cells and the action potential is propagated to the brain as sound sensation through cochlear nerve.

Defects of Ear

Deafness may be temporary or permanent. It can be further classified into **conductive deafness** and **sensory-neural deafness**. Possible causes for conductive deafness may be due to

i. the blockage of ear canal with earwax,
ii. Rupture of eardrum
iii. Middle ear infection with fluid accumulation
iv. Restriction of ossicular movement.

In **sensory-neural deafness**, the defect may be in the organ of Corti or the auditory nerve or in the ascending auditory pathways or auditory cortex.

Organ of Equilibrium

Balance is part of a sense called proprioception, which is the ability to sense

The intensity of sound is measured in decibels (dB). 0 dB is the threshold of hearing for normal ear. Severe hearing loss occurs with frequent or prolonged exposure to sound with intensities greater than 90dB. For meaningful conversations the intensity should be in the 50 dB range.